

Zia Hasan, PhD
Postdoctoral Fellow, Neurosciences

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Academic Background:

- **Memorial University of Newfoundland (MUN) | St. John's, NL, Canada**
 - Ph.D. in Neurosciences ([cGPA: 4.0/4.0](#)) | Courses: Systems Neuroscience, Biostatistics | Feb 2026
- **University of Rajshahi | Rajshahi, Bangladesh**
 - M.S. in Genetic Engineering & Biotechnology ([cGPA: 3.71/4.0](#)) | Dec 2017
 - B.Sc. in Genetic Engineering & Biotechnology ([cGPA: 3.75/4.0](#)) | Mar 2015

Core Competencies & Expertise

- **Bioinformatics & Computational Biology:** End-to-end R/Python pipelines for multi-omics data analysis · scRNA-seq · snRNA-seq · snATAC-seq · Whole-genome methylation analysis · HPC-based large-scale data processing.
- **Genomics & Wet Lab:** Extensive hands-on expertise in single-cell, single-nucleus, and bulk sequencing workflows · NGS library preparation · Nuclei isolation for multi-omics · Molecular validation techniques.
- **Molecular Biology & Diagnostics:** qPCR · Western blotting · Immunohistochemistry (IHC) · DNA/RNA isolation · FACS · PCR-based diagnostic assays for pathogen identification.
- **Microbial Biology:** Extensive experience in microbial culture · Biochemical and molecular characterization · Pathogen identification and characterization.
- **Data Curation & Reproducibility:** NCBI GEO (GSE272794) and GenBank (MG735450, MG948561, MH071191) submissions · Version-controlled and reproducible workflows using GitHub.
- **Creative & Technical Digital Skills:** Scientific figure design (Adobe Illustrator and Photoshop) · Web development (HTML/CSS/JavaScript) · iOS app development · Basic digital marketing tools.

Professional Experience

Academic & Research Positions:

- **Postdoctoral Fellow** | Dr. Qi Yuan lab, Neurosciences, Memorial University of Newfoundland, Canada (02/2026- Present)
 - Leading a spatial multi-omics (scRNA-seq and scATAC-seq) research program investigating cell-type-specific pathological signatures in neurodegenerative diseases.
 - Developing reproducible R/Python workflows and quality-control criteria for spatial multi-omics preprocessing and data integration.
 - Translating complex bioinformatics analyses into publication-quality visualizations and actionable insights for interdisciplinary collaborators.
- **PhD Researcher** | Dr. Qi Yuan & Dr. Touati Benoukraf labs, Neurosciences, Memorial University of Newfoundland, Canada (09/2019- 02/2026)
 - Designed and executed a comprehensive multi-cohort study investigating early Alzheimer's disease mechanisms using a novel pretangle tau rat model, integrating behavioral data with multi-omics profiling.
 - Established and optimized the laboratory's foundational single-nucleus genomics platform, leading to a widely accessed open-source protocol publication.
 - Published comprehensive analytical codebases and large-scale datasets (NCBI GEO: GSE272794), ensuring full reproducibility.

- **Visiting Researcher** | Dr. Kanae Ando Lab, Tokyo Metropolitan University, Japan (11/2022- 01/2023)
- Participated in specialized training and collaborative research that directly advanced the objectives of my PhD project.
- **Research Fellow** | Dr. Biswanath Sikdar, Ministry of Science and Technology, Bangladesh (2016 - 2018)
- Conducted microbial isolation and molecular characterization, resulting in the submission of three novel sequences to GenBank (MG735450, MG948561, MH071191).
- **Research Assistant (Part-time)** | Dr. Biswanath Sikdar, Joarder DNA & Chromosome Research Lab, University of Rajshahi, Bangladesh (2014 - 2016)
- Conducted plant pathology research using PCR and 16S/ITS sequencing for the identification of bacterial and fungal pathogens, including *Xanthomonas*, *Cercospora*, and *Colletotrichum*.

Industry Experience:

- **Executive, Research & Development** | Apex Biotechnology Laboratory, Bangladesh (11/2018-08/2019)
- Conducted pilot-scale fermentation · Bioreactor validation · PCR-based diagnostics · Protocol development and documentation to ensure process reproducibility.

Teaching & Service:

- **Executive Director of Communications** | Graduate Students' Union, Memorial University of Newfoundland, Canada (2021–2023)
- **Teaching Assistant (BIOL 1001 & BIOL 2041)** | Department of Biology, Memorial University of Newfoundland, Canada (2020–2021)

Fellowships & Awards

- **Fellow of the School of Graduate Studies** | *Graduate Studies, MUN* | 2025-2026
- **Best PhD Oral Presentation (3rd, Neurosciences)** | **CA\$125** | *Faculty of Medicine, MUN* | 2025
- **Apu Akbar Memorial Award** | **CA\$500** | *School of Graduate Studies, MUN* | 2023
- **ARC-NL Fellowship** | **CA\$20,000** | *Aging Research Centre, NL* | 2020-2022
- **Dean's Building a Healthy Tomorrow Award** | **CA\$2,544** | *Faculty of Medicine, MUN* | 2021-2022
- **Dr. Angus J. Neary Genetics Scholarship** | **CA\$2,831** | *Faculty of Medicine, MUN* | 2020-2021
- **Dean's Fellowship Award** | **CA\$80,000** | *Faculty of Medicine, MUN* | 2019-2023
- **NST Fellowship** | **CA\$1000** | *Ministry of Science and Technology, Bangladesh* | 2016-2017
- **Talent Scholarship (Tuition Waiver)** | *University of Rajshahi, Bangladesh* | 2015-2016

Publications (Selected)

I. Articles published in peer-reviewed journals

- [1] **Hasan, Z.**, Torraville, S. E., Omoluabi, T., Maziar, A., Belolise, O. N., MacGowan, L. A., Flynn, C. M., Reinhardt, C., Walling, S. G., Benoukraf, T., & Yuan, Q. (2025). *Sex and life experience shape locus coeruleus pretangle tau pathology*. *Alzheimer's & Dementia*, 22(3), e71285. <https://doi.org/10.1002/alz.71285>
- [2] Rae, R. J., Alberhasky, J. M. H., Baillet, M., Bangasser, D. A., Belloy, M. E., Berry, A. S., ...**Hasan, Z.**,... & Kelberman, M. A. (2026). Sex differences in neuromodulatory subcortical systems and their implications for Alzheimer's disease. *Alzheimer's & dementia*, 22(3), e71291. <https://doi.org/10.1002/alz.71291>

- [3] Sepahvand, T., Qin, T., **Hasan, Z.**, Power, K. D., Carew, S. J., Nazari, N., Janes, A. M., et al. (2025). *Second-order threat conditioning in the amygdala–posterior piriform cortex network*. *Communications Biology*, 8(1), 1–18. <https://doi.org/10.1038/s42003-025-08287-2>
- [4] Omoluabi, T. (†), **Hasan, Z.** (†), Piche, J. E. (†), Flynn, A. R. S., Doré, J. J. E., Walling, S. G., et al. (2025). *Locus coeruleus vulnerability to tau hyperphosphorylation in a rat model*. *Aging Cell*, 24(3), e14405. <https://doi.org/10.1111/acer.14405> † Co–first authors
- [5] Crossley, Q. Y. C., Omoluabi, T., Torraville, S. E., **Hasan, Z.**, et al. (2024). *Hippocampal hyperphosphorylated tau-induced deficiency is rescued by L-type calcium channel blockade*. *Brain Communications*, 6(2). <https://doi.org/10.1093/braincomms/fcae096>

II. Non-peer-reviewed articles/presentations

- [1] **Hasan, Z.**, Benoukraf, T., & Yuan, Q. (2026). *Updating the Reference Genome from mRatBN7.2 to GRCr8 Improves Single-Nucleus RNA-Sequencing Sensitivity in the Rat Brain* (Manuscript in Preparation).
- [2] **Hasan, Z.**, Benoukraf, T., & Yuan, Q. (2026). Sex and Life Experience Converge on Hippocampal Transcriptomic Responses in a Locus Coeruleus Pretangle Tau Model *Alzheimer's Association International Conference (AAIC)*; London, UK.
- [3] **Hasan, Z.**, Benoukraf, T., & Yuan, Q. (2025). Nuclei Isolation Protocol from Fresh and Frozen Tissue for Single-Nucleus RNA-Seq and ATAC-Seq. *protocols.io*. <https://dx.doi.org/10.17504/protocols.io.14egn228zg5d/v1>
- [4] **Hasan, Z.**, Torraville, S.E., Reinhardt, C., Benoukraf, T., & Yuan, Q. (July 2025). Life Experience Tunes Cell Type-Specific Transcriptomic Responses to Early Tau Pathology. *BioMedicine Graduate Symposium*; Memorial University, St. John's (NL), Canada.
- [5] **Hasan, Z.**, Torraville, S. E., Omoluabi, T., Maziar, A., Belolise, O. N., MacGowan, L. A., Flynn, C. M., Benoukraf, T., & Yuan, Q. (July 2025). Environmental modulation of early tau pathology in Alzheimer's disease: A pretangle-tau rat model. *Alzheimer's Association International Conference (AAIC)*; Toronto (ON), Canada.
- [6] **Hasan, Z.**, Torraville, S.E., Reinhardt, C., Benoukraf, T., & Yuan, Q. (May 2025). Postnatal stress exacerbates, while enrichment mitigates, locus coeruleus pretangle tau-induced gene alterations in the hippocampus. *Canadian Neuroscience Meeting (CAN 2025)*; Toronto (ON), Canada.
- [7] **Hasan, Z.**, Omoluabi, T., Piche, J. E., et al. (2025). Locus coeruleus vulnerability to tau hyperphosphorylation in a rat model. NCBI Gene Expression Omnibus (GEO). Accession: GSE272794.
- [8] **Hasan, Z.**, Torraville, S.E., Reinhardt, C., Benoukraf, T., & Yuan, Q. (November 2023). The effect of environmental factors in a rat pretangle tau model of Alzheimer's disease: A single-nuclei transcriptomics study. *Society for Neuroscience (SfN) Annual Meeting*; Washington (DC), USA.
- [9] **Hasan, Z.** (July 2023). The effects of early life events in a rat pretangle tau model: a single-nuclei transcriptomics study. *BioMedicine Graduate Symposium*; Memorial University, St. John's (NL), Canada.